

Rig Drill Sensors

Wiring and calibrating the rig sensors an MWD system depends on

⚠ FOR UNDERSTANDING PURPOSES ONLY

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Flutter / Dart

Hook load

Drawworks

Pump strokes

Standpipe pressure

1. What It Is

An MWD system needs to know what the rig is doing: how heavy the string is, where the block is, how fast the pumps are running and what the standpipe pressure is. The sensors that measure this must be **wired and calibrated** before a job. This module simulates that, in the same hands-on spirit as the other MWD tools. It is reached from Operations › LWD / MWD › MWD & DD › Rig Sensor.

The opening screen shows two large cards, one for each page of sensors.

2. Page 1 — Hook Load & Drawworks

Three tabs:

Tab	What you do
Hook Load	See the deadline anchor picture and a wiring diagram with analog and digital connection branches. Fill in a calibration table of six rows (no drill string, then 30, 60, 90, 120 and 150 stands), pairing a sensor voltage with a hook load in pounds, against the top-drive system weight. A Reset button clears it
Drawworks	See the drawworks picture and wiring. Calibrate the sensor count against height in metres, then test it with a top-drive simulator using Hoist and Slack buttons
Dashboard	A live view: bit depth, a depth-against-time chart and controls to Connect Stand and Disconnect DP . A threshold weight and the current hook load decide when a connection is recognised. Stand details and a total stands summary build up as you go

3. Page 2 — Pump Strokes & Standpipe Pressure

Another set of tabs for the pump and pressure side:

- **Pump stroke sensor** — a mud pump read-out with total strokes, a **Reset Strokes** button, pump output and pump GPM.
- **SPP sensor** — the standpipe pressure gauge.
- **Pulse / waveform and MWD valve / orifice** — visualisations of the pressure pulse the MWD tool creates.
- **Fix GPM values** — sets the flow figures used by the simulation.

Sensors must be wired first

If the sensors have not been plugged in, the screen shows “Sensors not wired” with a “Go to wiring” shortcut. This reflects the real order of work: nothing downstream means anything until the sensors are connected.

A **Reset All** button returns the whole page to its starting state.

4. Calibration Data Behaviour

- The no-drill-string row fills in its voltage by itself, as a read-only value.
- Other rows accept entries within a realistic voltage range.
- The screen is split into a fixed header that never scrolls and content that does, so the tabs stay reachable.